

Can soy phytoestrogens decrease DNA methylation in BRCA1 and BRCA2 oncosuppressor genes in breast cancer?

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Although soy phytoestrogens have been postulated to exert a protective effect against breast cancer, the attendant mechanisms, in particular epigenetics underpinnings, have remained elusive. We investigated the putative effects on DNA methylation by two naturally occurring isoflavones, genistein and daidzein, in a study of the BRCA1 and BRCA2 oncosuppressor genes in breast cancer cell lines (MCF-7, MDA-MB 231, and MCF10a). A demethylant agent, the 5-azacytidine, and a methylant, the budesonide, were used as treatment controls. DNA methylation of BRCA1 and BRCA2 was investigated with methylated DNA immunoprecipitation coupled with PCR. In parallel, protein expression was determined by Western blot, immunohistochemistry, and confocal microscopy. Our results suggest that treatment with 18.5 μ M Genistein or 78.5 μ M Daidzein might reverse DNA hypermethylation and restore the expression of the oncosuppressor genes BRCA1 and BRCA2. 5-Azacytidine also enhanced the reexpression of these genes while budesonide had an opposite effect. To the best of our knowledge, these observations, while requiring replication, provide new evidence on potential epigenetic mechanisms by which genistein and daidzein might contribute to regulation of the BRCA1 and BRCA2. Future studies are warranted on whether the demethylating effect of genistein and daidzein is global or focused on select candidate genes.